LOCALLY OF FINITE TYPE MORPHISMS ARE ETALE-LOCALLY FINITE OVER SMOOTH MAPS WITH GEOMETRICALLY INTEGRAL FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 13431-13460 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 13462-13520, the complete author proof (57 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The schemes-side slogan (finite type is etale-locally finite over smooth) is retained verbatim in the statement span; this is the refined local-local version, distinct from the plain local structure lemma it builds on.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The chapter's local structure lemma for finite type morphisms (lemma-local-structure-finite-type), existence of elementary etale neighbourhoods trivialising fibres with geometrically irreducible fibres, and standard properties of smooth maps (quasi-compact separated smooth over an affine base is of finite presentation); cited inputs with the diagram construction written in the retained span.

Source reference. Original lemma 052E: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13383–13402. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 057J: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13327–13336. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 056W: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5553–5555. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 033M: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13520–13520. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 04KV: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13520–13520. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 056T: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 13520–13520. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *A morphism of finite type is, in étale neighbourhoods, finite over a smooth morphism. Let $f : X \to S$ be a morphism. Let $x \in X$ and set $s = f(x)$. Assume that f is locally of finite type and that $n = \dim_x(X_s)$. Then there exists a commutative diagram*

$$\begin{array}{ccc} X & \xleftarrow{g} & X' \\ \downarrow & & \downarrow{\scriptstyle \pi} \\ & & Y' \\ & & \downarrow{\scriptstyle h} \\ S & \xleftarrow{e} & S' \end{array} \qquad \begin{array}{ccc} x & \longleftarrow & x' \\ \downarrow & & \downarrow \\ & & y' \\ & & \downarrow \\ s & \longleftarrow & s' \end{array}$$

and a point $x' \in X'$ with $g(x') = x$ such that with $y' = \pi(x')$, $s' = h(y')$ we have

1. *$h : Y' \to S'$ is smooth of relative dimension n,*
2. *all fibres of $Y' \to S'$ are geometrically integral,*
3. *$g : (X', x') \to (X, x)$ is an elementary étale neighbourhood,*
4. *π is finite, and $\pi^{-1}(\{y'\}) = \{x'\}$,*
5. *$\kappa(y')$ is a purely transcendental extension of $\kappa(s')$, and*
6. *$e : (S', s') \to (S, s)$ is an elementary étale neighbourhood.*

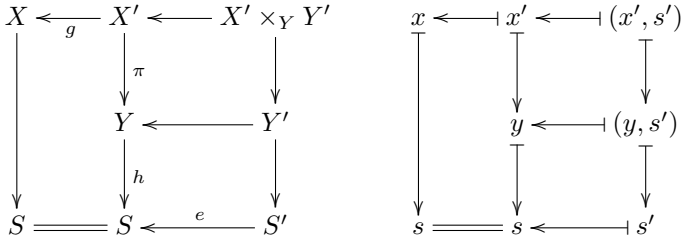
Moreover, if f is locally of finite presentation, then π is of finite presentation.

Proof. The question is local on S, hence we may replace S by an affine open neighbourhood of s. Next, we apply Lemma 052E to get a commutative diagram

$$\begin{array}{ccc} X & \xleftarrow{g} & X' \\ \downarrow & & \downarrow{\scriptstyle \pi} \\ & & Y \\ & & \downarrow{\scriptstyle h} \\ S & = & S \end{array} \qquad \begin{array}{ccc} x & \longleftarrow & x' \\ \downarrow & & \downarrow \\ & & y \\ & & \downarrow \\ s & = & s \end{array}$$

where h is smooth of relative dimension n and $\kappa(y)$ is a purely transcendental extension of $\kappa(s)$. Since the question is local on X also, we may replace Y by an affine neighbourhood of y (and X' by the inverse image of this under π). As S is affine this guarantees that $Y \to S$ is quasi-compact, separated and smooth, in particular of finite presentation. Let T be the connected component of Y_s containing y. As Y_s is Noetherian we see that T is open. We also see that T is geometrically connected over $\kappa(s)$ by Varieties, Lemma 04KV. Since T is also smooth over $\kappa(s)$ it is geometrically normal, see Varieties, Lemma 056T. We conclude that T is geometrically irreducible over $\kappa(s)$ (as a connected Noetherian normal scheme is irreducible, see Properties, Lemma 033M). Finally, note that the smooth morphism h is normal by Lemma 056W. At this point we have verified all assumption of Lemma 057J hold for the morphism $h : Y \to S$ and open $T \subset Y_s$. As a result of

applying Lemma 057J we obtain $e : S' \to S$, $s' \in S'$, Y' as in the commutative diagram



where $e : (S', s') \to (S, s)$ is an elementary étale neighbourhood, and where $Y' \subset Y_{S'}$ is an open neighbourhood all of whose fibres over S' are geometrically irreducible, such that $Y'_{s'} = T$ via the identification $Y_s = Y_{S',s'}$. Let $(y, s') \in Y'$ be the point corresponding to $y \in T$; this is also the unique point of $Y \times_S S'$ lying over y with residue field equal to $\kappa(y)$ which maps to s' in S'. Similarly, let $(x', s') \in X' \times_Y Y' \subset X' \times_S S'$ be the unique point over x' with residue field equal to $\kappa(x')$ lying over s'. Then the outer part of this diagram is a solution to the problem posed in the lemma. Some minor details omitted. □